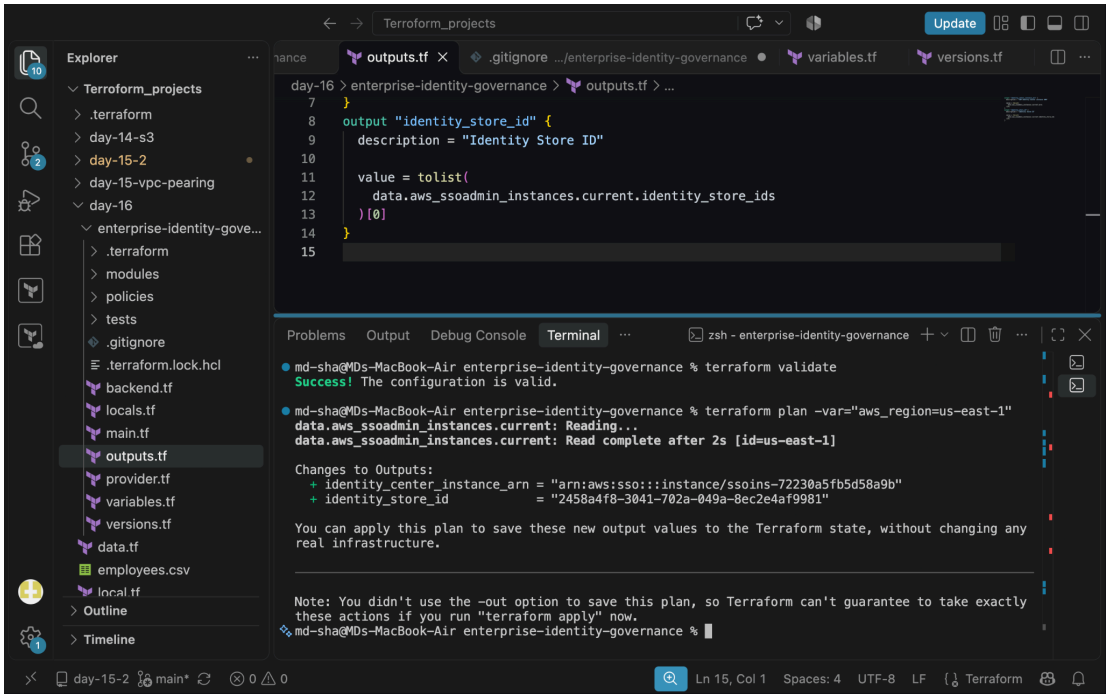


hcl

```hcl
csvdecode(file("employees.csv"))
```

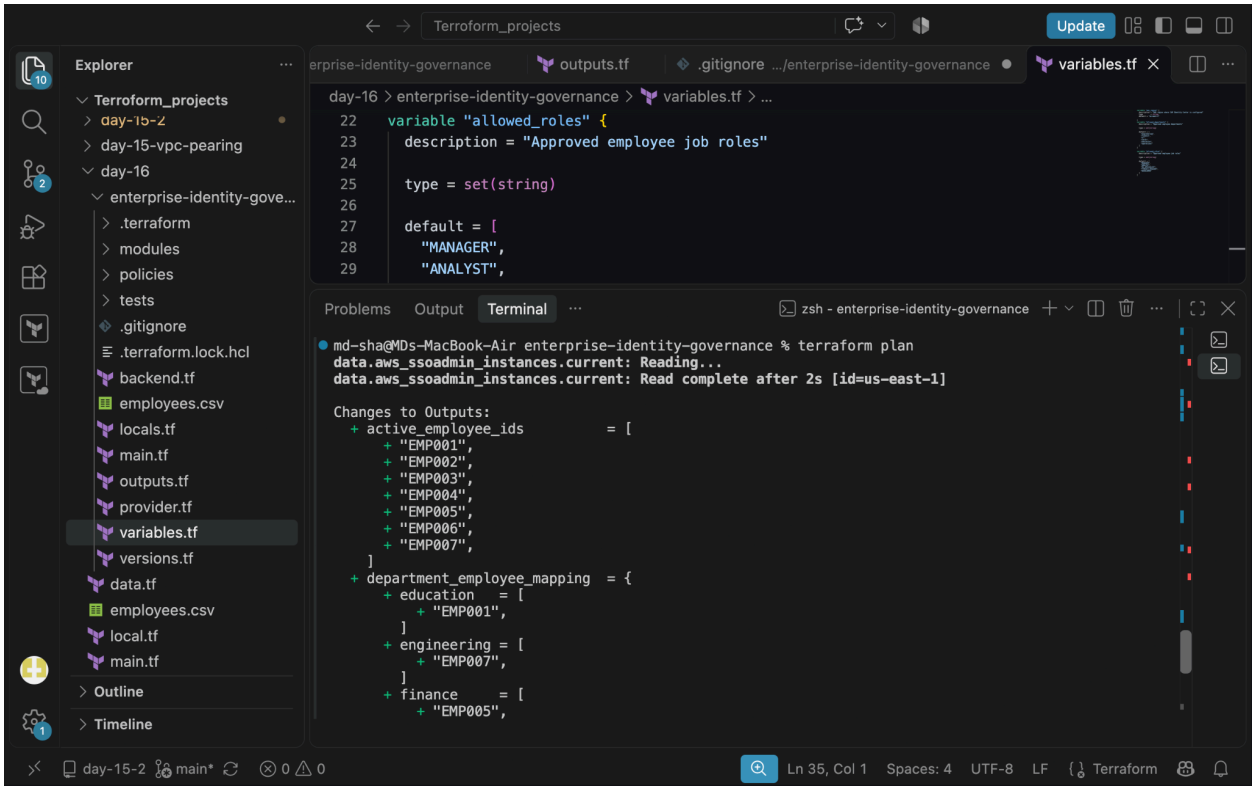
Two functions are working together:

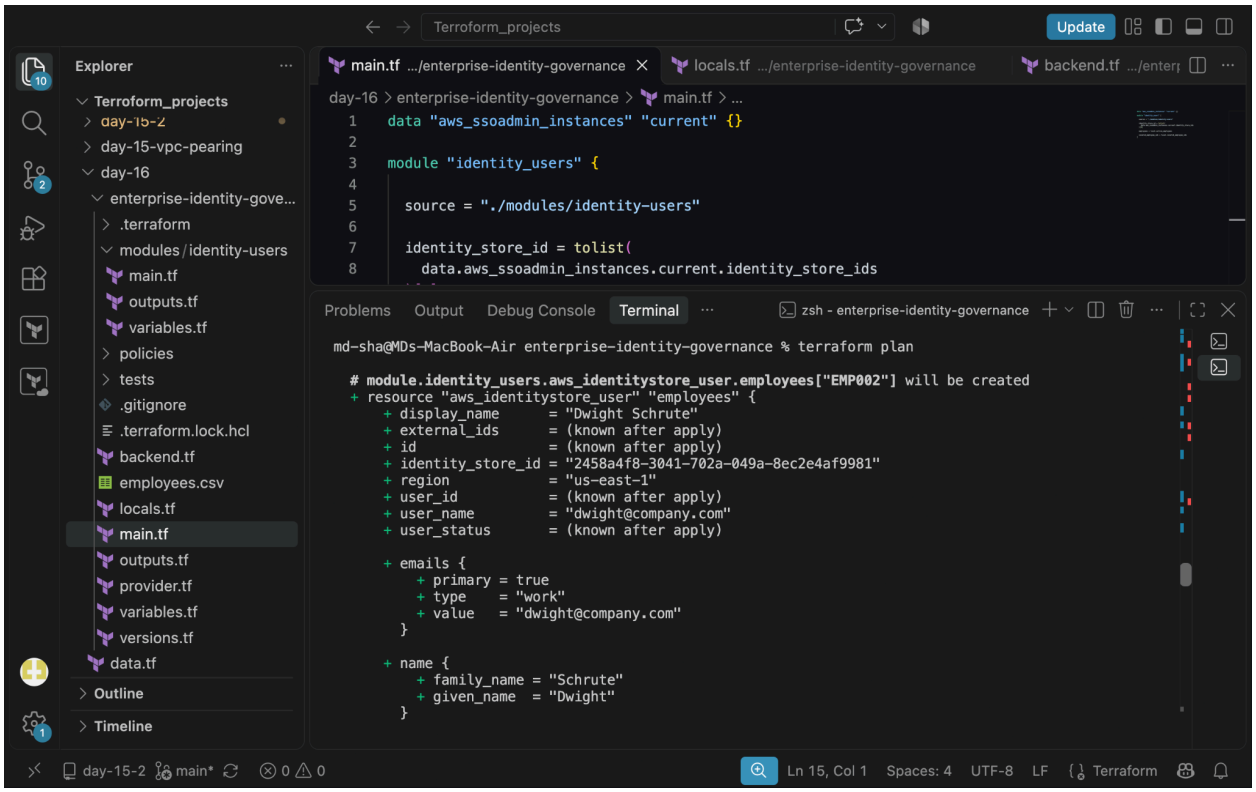
- `file()` reads the CSV file.
- `csvdecode()` converts the CSV contents into a list of maps.

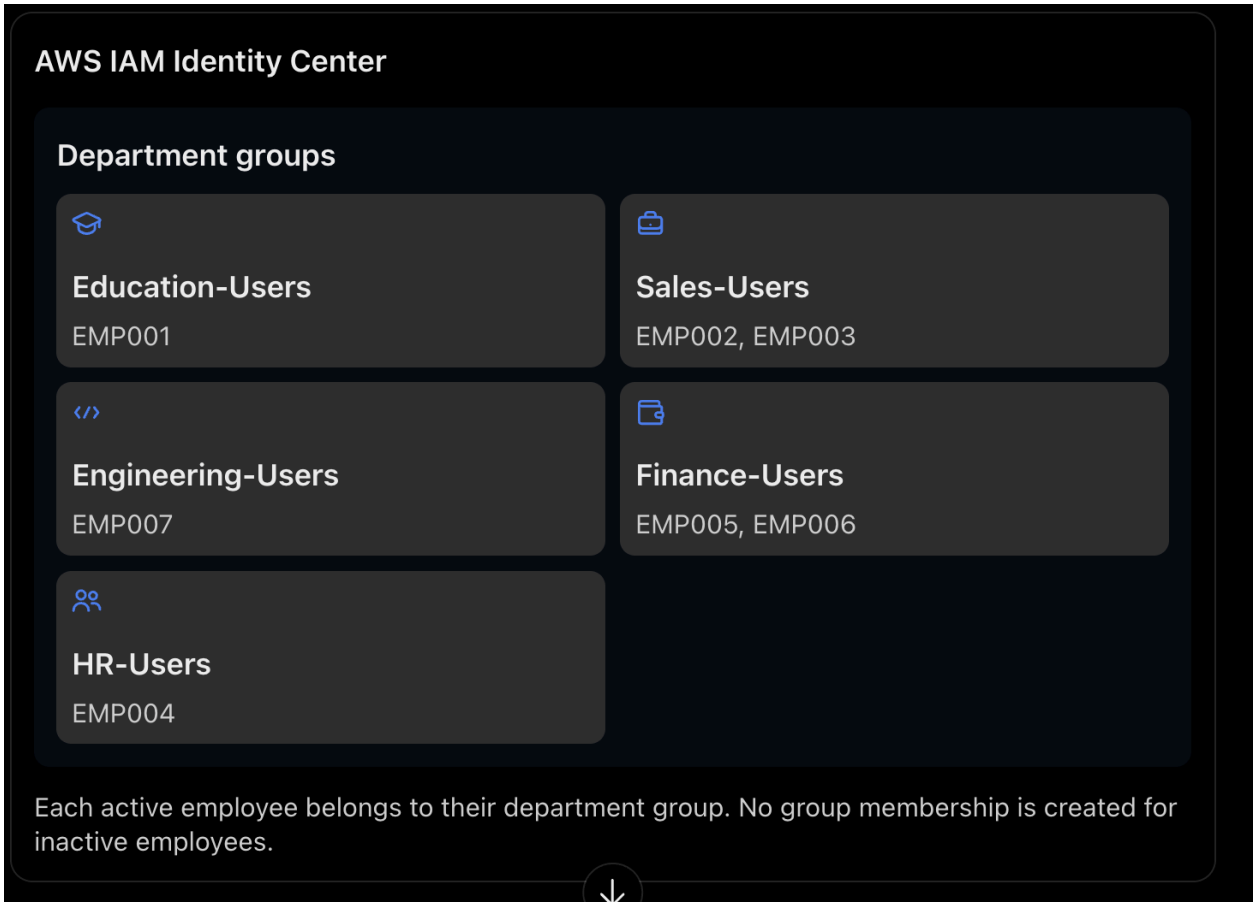
Conceptually, Terraform converts the CSV into something like:

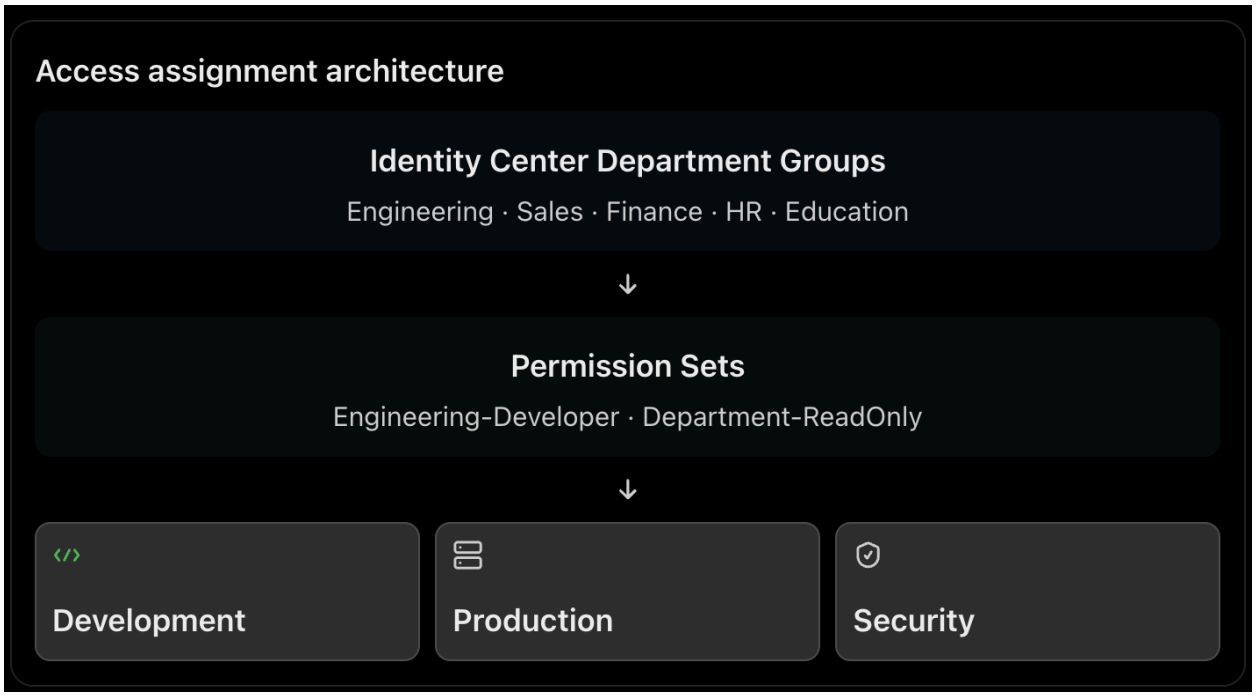
hcl

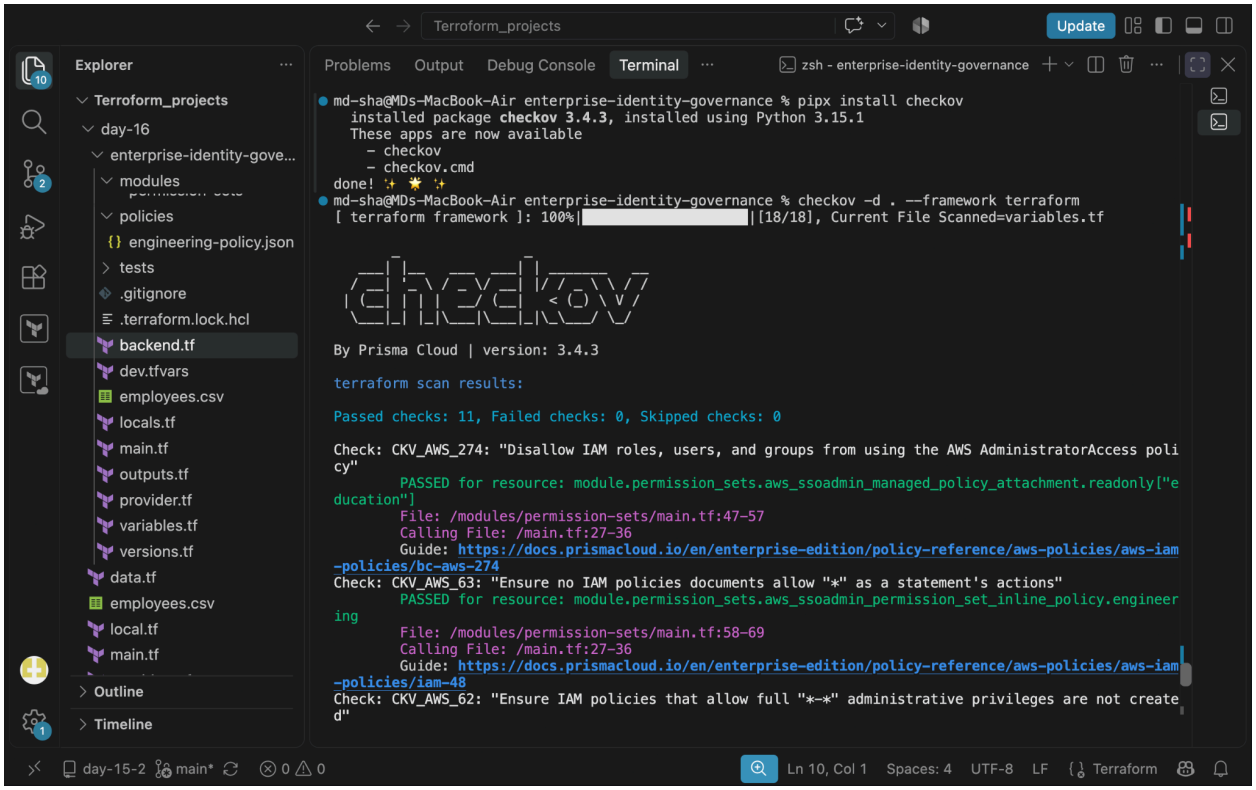
```hcl
[
  {
    employee_id      = "EMP001"
    first_name       = "Michael"
    last_name        = "Scott"
    department       = "Education"
    role             = "MANAGER"
    employment_status = "ACTIVE"
  },
  {
    employee_id      = "EMP002"
    first_name       = "Dwight"
    last_name        = "Schrute"
    department       = "Sales"
    role             = "ANALYST"
    employment_status = "ACTIVE"
  }
]
```

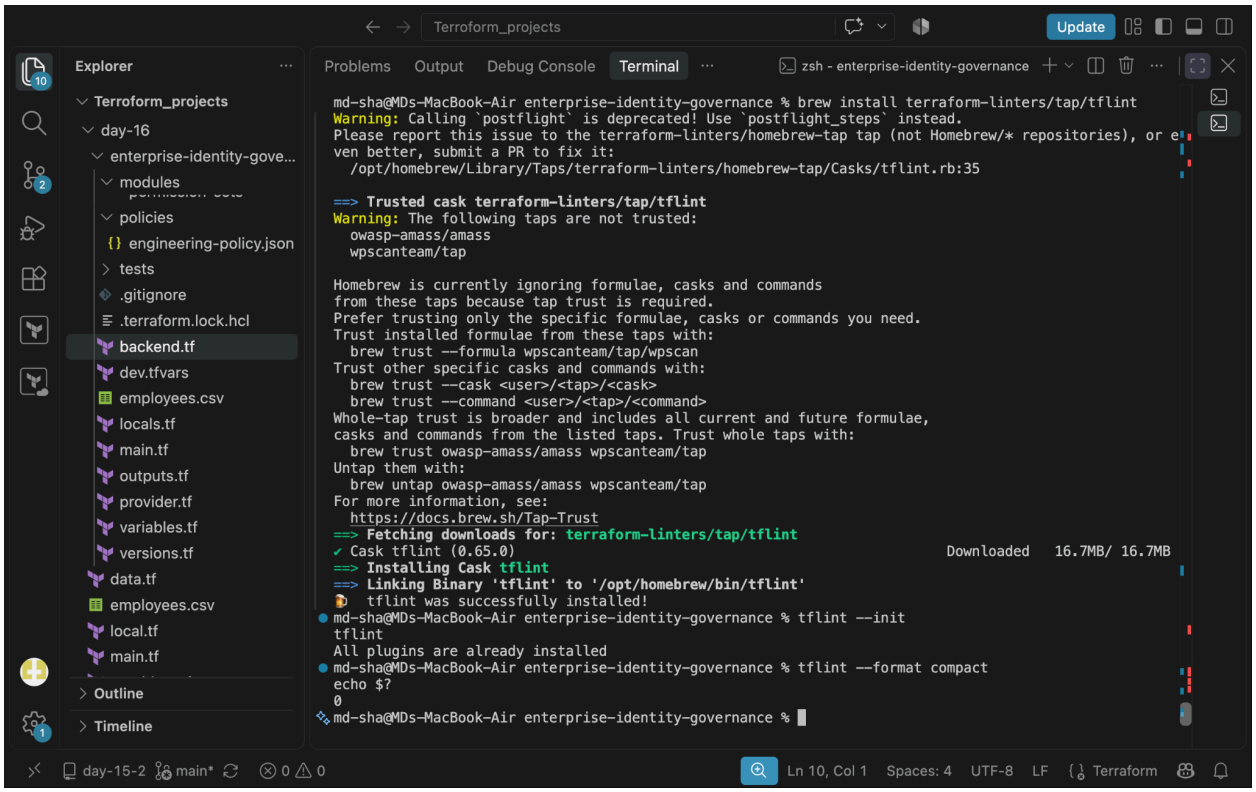

Terroform_projects
Update
Explorer
Terroform_projects
day-15-2
day-15-vpc-pearing
day-16
enterprise-identity-gove...
.terraform
modules
policies
tests
.gitignore
.terraform.lock.hcl
backend.tf
employees.csv
locals.tf
main.tf
outputs.tf
provider.tf
variables.tf
versions.tf
data.tf
employees.csv
local.tf
main.tf
Outline
Timeline
outputs.tf
.gitignore .../enterprise-identity-governance
variables.tf
day-16 > enterprise-identity-governance > variables.tf > ...
22 variable "allowed_roles" {
23 description = "Approved employee job roles"
25 type = set(string)
27 default = [
28 "MANAGER",
29 "ANALYST",
Problems
Output
Terminal
zsh - enterprise-identity-governance
md-sha@MDs-MacBook-Air enterprise-identity-governance % terraform plan
data.aws_ssoadmin_instances.current: Reading...
data.aws_ssoadmin_instances.current: Read complete after 2s [id=us-east-1]
Changes to Outputs:
+ active_employee_ids = [
+ "EMP001",
+ "EMP002",
+ "EMP003",
+ "EMP004",
+ "EMP005",
+ "EMP006",
+ "EMP007",
]
+ department_employee_mapping = {
+ education = [
+ "EMP001",
]
+ engineering = [
+ "EMP007",
]
+ finance = [
+ "EMP005",
day-15-2
main*
0 0
Ln 35, Col 1
Spaces: 4
UTF-8
LF
Terraform

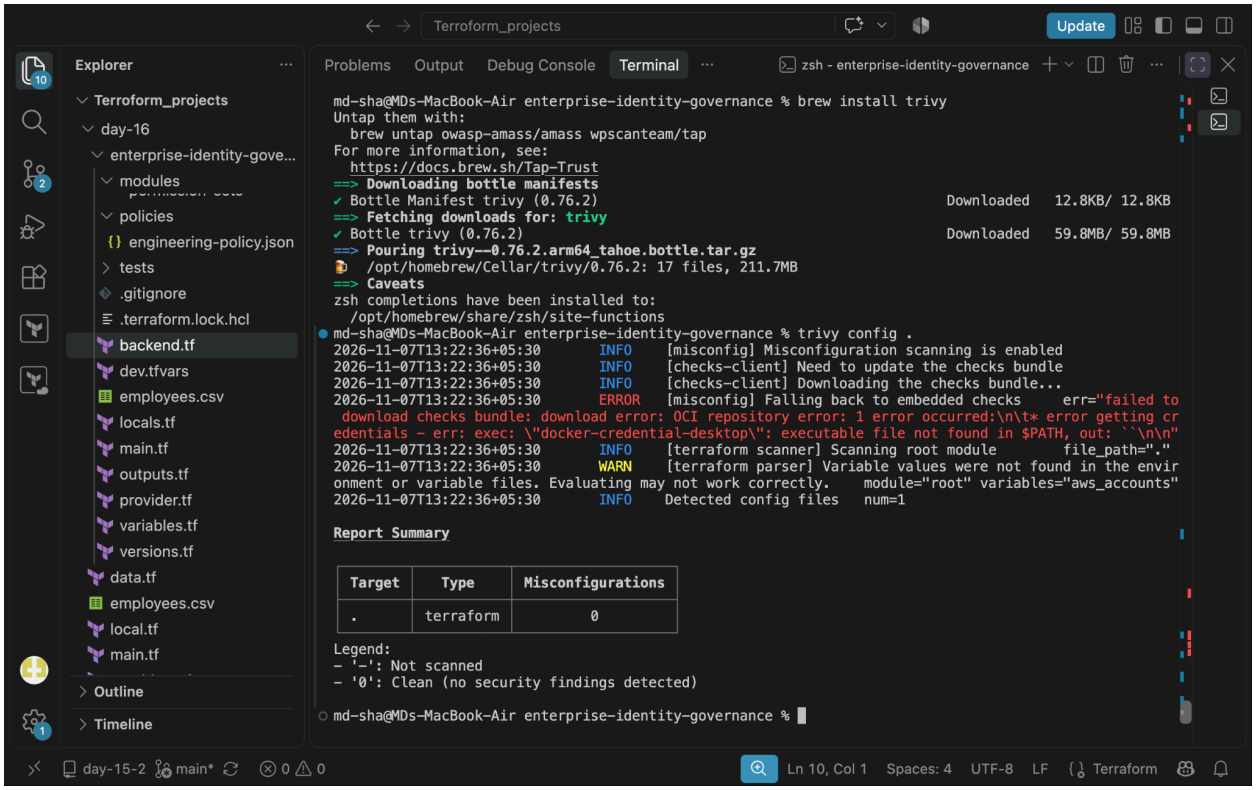

Terroform_projects
Update
Explorer
Terroform_projects
day-15-2
day-15-vpc-pearing
day-16
enterprise-identity-gove...
.terraform
modules / identity-users
main.tf
outputs.tf
variables.tf
policies
tests
.gitignore
.terraform.lock.hcl
backend.tf
employees.csv
locals.tf
main.tf
outputs.tf
provider.tf
variables.tf
versions.tf
data.tf
Outline
Timeline
main.tf .../enterprise-identity-governance
locals.tf .../enterprise-identity-governance
backend.tf .../enter
day-16 > enterprise-identity-governance > main.tf > ...
1 data "aws_ssoadmin_instances" "current" {}
3 module "identity_users" {
5 source = "./modules/identity-users"
7 identity_store_id = tolist(
8 data.aws_ssoadmin_instances.current.identity_store_ids
Problems
Output
Debug Console
Terminal
zsh - enterprise-identity-governance
md-sha@MDs-MacBook-Air enterprise-identity-governance % terraform plan
module.identity_users.aws_identitystore_user.employees["EMP002"] will be created
+ resource "aws_identitystore_user" "employees" {
+ display_name = "Dwight Schrute"
+ external_ids = (known after apply)
+ id = (known after apply)
+ identity_store_id = "2458a4f8-3041-702a-049a-8ec2e4af9981"
+ region = "us-east-1"
+ user_id = (known after apply)
+ user_name = "dwight@company.com"
+ user_status = (known after apply)
+ emails {
+ primary = true
+ type = "work"
+ value = "dwight@company.com"
}
+ name {
+ family_name = "Schrute"
+ given_name = "Dwight"
}
day-15-2
main*
0 0
Ln 15, Col 1
Spaces: 4
UTF-8
LF
Terraform

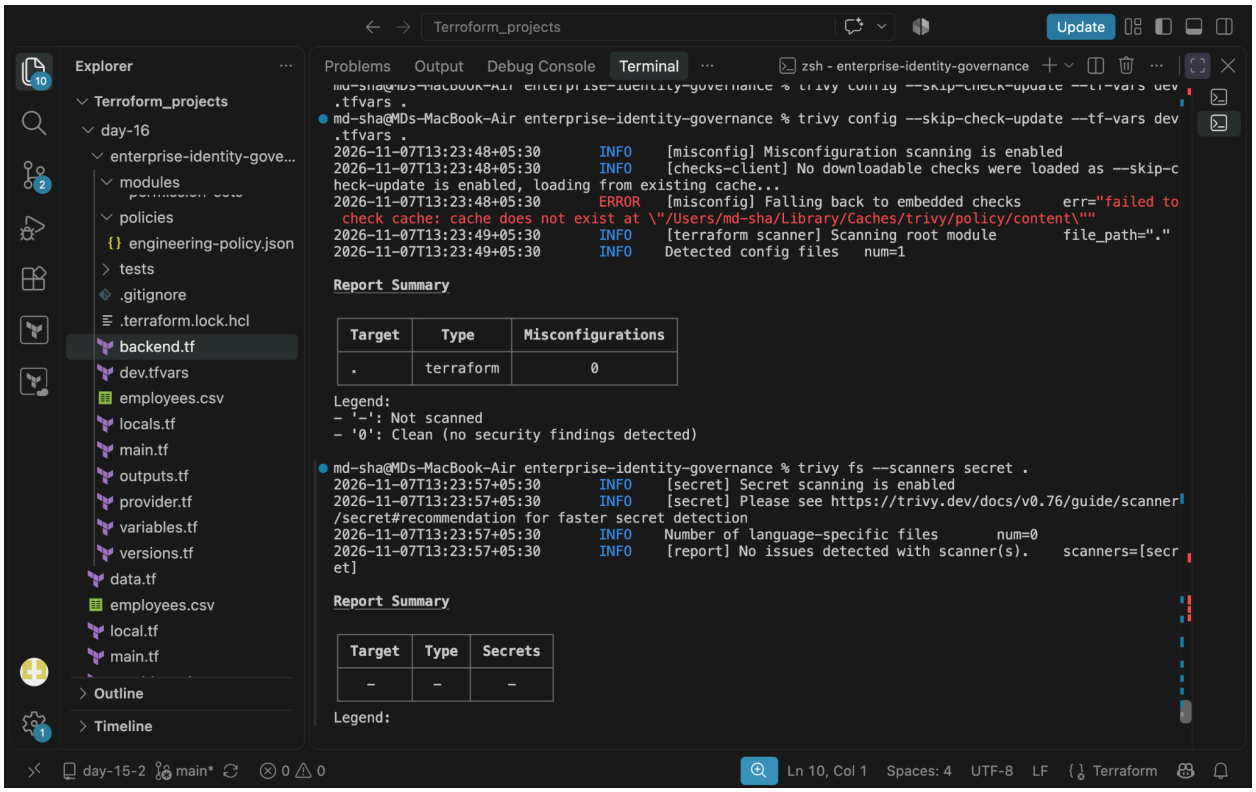

AWS IAM Identity Center
Department groups
Education-Users
EMP001
Sales-Users
EMP002, EMP003
Engineering-Users
EMP007
Finance-Users
EMP005, EMP006
HR-Users
EMP004
Each active employee belongs to their department group. No group membership is created for inactive employees.

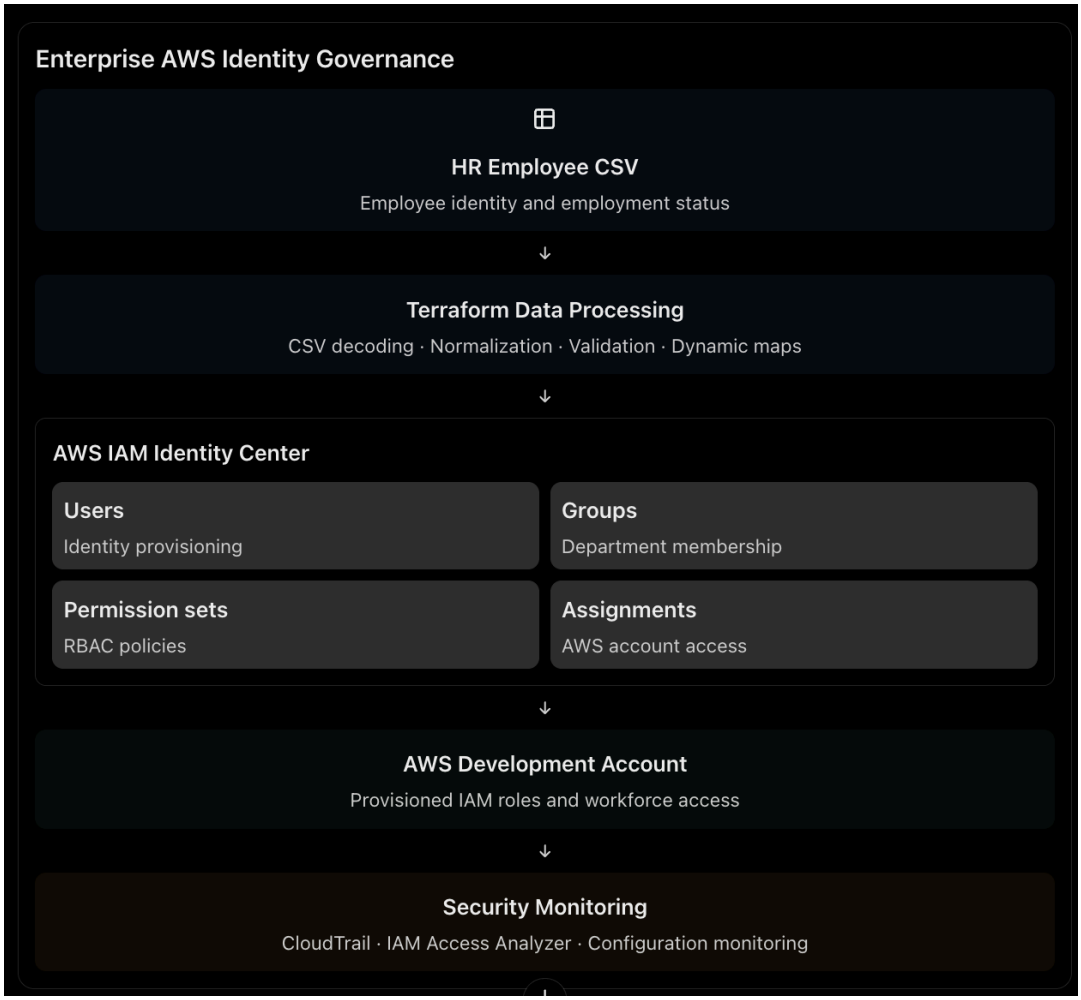

Access assignment architecture
Identity Center Department Groups
Engineering · Sales · Finance · HR · Education
Permission Sets
Engineering-Developer · Department-ReadOnly
Development
Production
Security

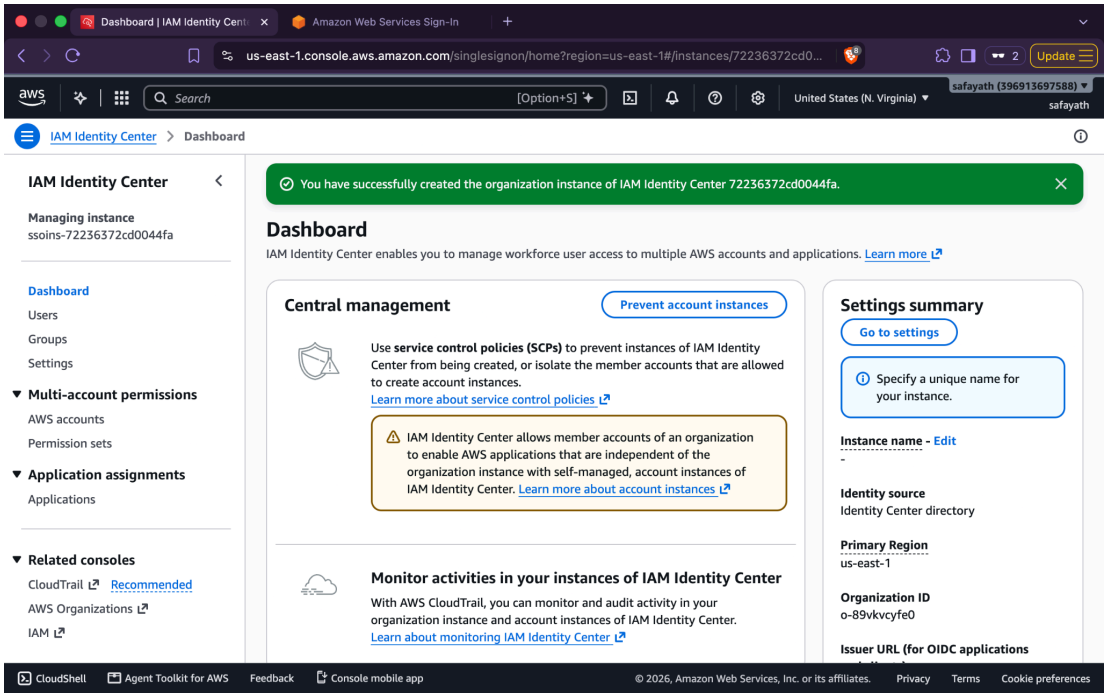

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locals.tf
main.tf
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provider.tf
variables.tf
versions.tf
data.tf
employees.csv
local.tf
main.tf
Outline
Timeline
md-sha@MDs-MacBook-Air enterprise-identity-governance % pipx install checkov
installed package checkov 3.4.3, installed using Python 3.15.1
These apps are now available
- checkov
- checkov.cmd
done! ✨ 🌟 ✨
md-sha@MDs-MacBook-Air enterprise-identity-governance % checkov -d . --framework terraform
[terraform framework]: 100%|[18/18], Current File Scanned=variables.tf
By Prisma Cloud | version: 3.4.3
terraform scan results:
Passed checks: 11, Failed checks: 0, Skipped checks: 0
Check: CKV_AWS_274: "Disallow IAM roles, users, and groups from using the AWS AdministratorAccess policy"
PASSED for resource: module.permission_sets.aws_ssoadmin_managed_policy_attachment.readonly["education"]
File: /modules/permission-sets/main.tf:47-57
Calling File: /main.tf:27-36
Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-iam-policies/bc-aws-274
Check: CKV_AWS_63: "Ensure no IAM policies documents allow "*" as a statement's actions"
PASSED for resource: module.permission_sets.aws_ssoadmin_permission_set_inline_policy.engineering
File: /modules/permission-sets/main.tf:58-69
Calling File: /main.tf:27-36
Guide: https://docs.prismacloud.io/en/enterprise-edition/policy-reference/aws-policies/aws-iam-policies/iam-48
Check: CKV_AWS_62: "Ensure IAM policies that allow full "*-*" administrative privileges are not created"
day-15-2
main*
0 0
Ln 10, Col 1
Spaces: 4
UTF-8
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Terraform


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engineering-policy.json
tests
.gitignore
.terraform.lock.hcl
backend.tf
dev.tfvars
employees.csv
locals.tf
main.tf
outputs.tf
provider.tf
variables.tf
versions.tf
data.tf
employees.csv
local.tf
main.tf
Outline
Timeline
md-sha@MDs-MacBook-Air enterprise-identity-governance % brew install terraform-linters/tap/tflint
Warning: Calling `postflight` is deprecated! Use `postflight_steps` instead.
Please report this issue to the terraform-linters/homebrew-tap tap (not Homebrew/* repositories), or even better, submit a PR to fix it:
/opt/homebrew/Library/Taps/terraform-linters/homebrew-tap/Casks/tflint.rb:35
==> Trusted cask terraform-linters/tap/tflint
Warning: The following taps are not trusted:
owasp-amass/amass
wpscanteam/tap
Homebrew is currently ignoring formulae, casks and commands
from these taps because tap trust is required.
Prefer trusting only the specific formulae, casks or commands you need.
Trust installed formulae from these taps with:
brew trust --formula wpscanteam/tap/wpscan
Trust other specific casks and commands with:
brew trust --cask <user>/<tap>/<cask>
brew trust --command <user>/<tap>/<command>
Whole-tap trust is broader and includes all current and future formulae,
casks and commands from the listed taps. Trust whole taps with:
brew trust owasp-amass/amass wpscanteam/tap
Untap them with:
brew untap owasp-amass/amass wpscanteam/tap
For more information, see:
https://docs.brew.sh/Tap-Trust
==> Fetching downloads for: terraform-linters/tap/tflint
✔ Cask tflint (0.65.0) Downloaded 16.7MB/ 16.7MB
==> Installing Cask tflint
==> Linking Binary 'tflint' to '/opt/homebrew/bin/tflint'
tflint was successfully installed!
md-sha@MDs-MacBook-Air enterprise-identity-governance % tflint --init
tflint
All plugins are already installed
md-sha@MDs-MacBook-Air enterprise-identity-governance % tflint --format compact
echo $?
0
md-sha@MDs-MacBook-Air enterprise-identity-governance %
day-15-2
main*
0 0
Ln 10, Col 1
Spaces: 4
UTF-8
LF
Terraform

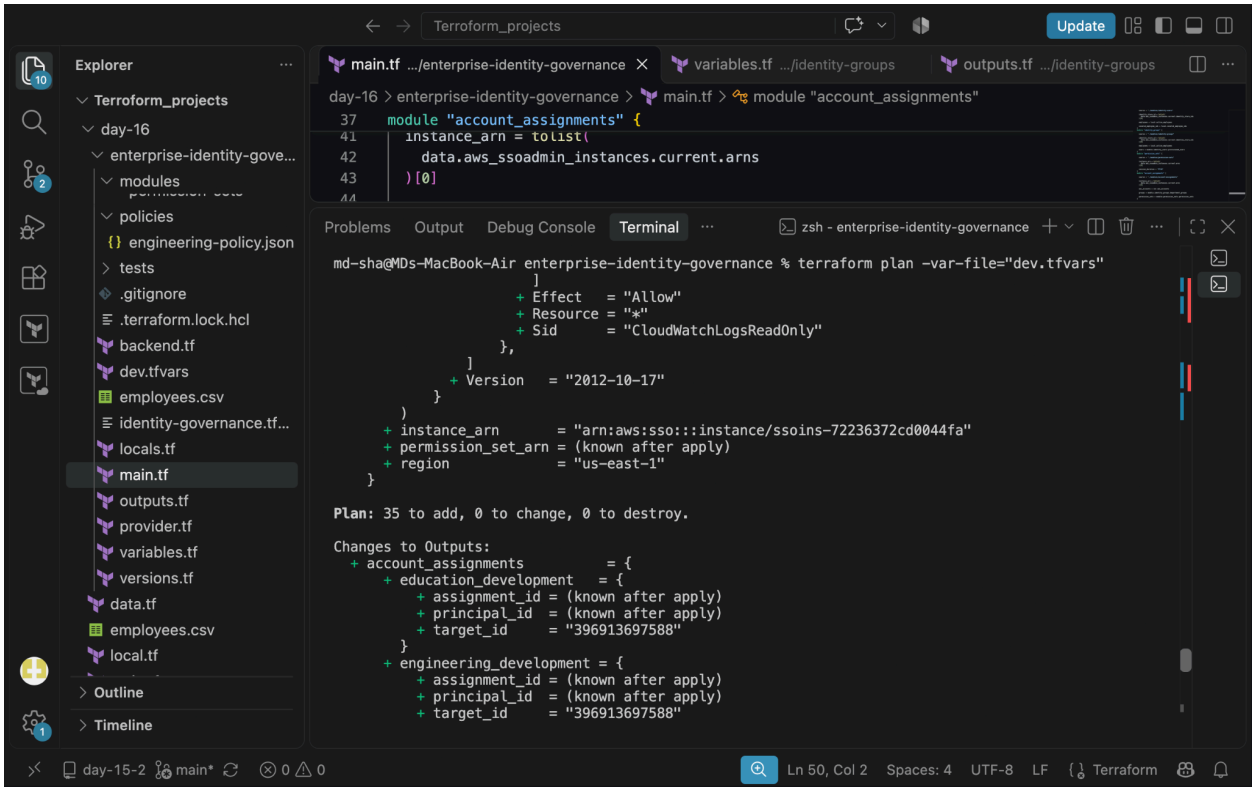

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modules
policies
engineering-policy.json
tests
.gitignore
.terraform.lock.hcl
backend.tf
dev.tfvars
employees.csv
locals.tf
main.tf
outputs.tf
provider.tf
variables.tf
versions.tf
data.tf
employees.csv
local.tf
main.tf
Outline
Timeline
Problems
Output
Debug Console
Terminal
zsh - enterprise-identity-governance
md-sha@MDs-MacBook-Air enterprise-identity-governance % brew install trivy
Untap them with:
brew untap owasp-amass/amass wpscanteam/tap
For more information, see:
https://docs.brew.sh/Tap-Trust
==> Downloading bottle manifests
✔ Bottle Manifest trivy (0.76.2) Downloaded 12.8KB/ 12.8KB
==> Fetching downloads for: trivy
✔ Bottle trivy (0.76.2) Downloaded 59.8MB/ 59.8MB
==> Pouring trivy--0.76.2.arm64_tahoe.bottle.tar.gz
/opt/homebrew/Cellar/trivy/0.76.2: 17 files, 211.7MB
==> Caveats
zsh completions have been installed to:
/opt/homebrew/share/zsh/site-functions
md-sha@MDs-MacBook-Air enterprise-identity-governance % trivy config .
2026-11-07T13:22:36+05:30 INFO [misconfig] Misconfiguration scanning is enabled
2026-11-07T13:22:36+05:30 INFO [checks-client] Need to update the checks bundle
2026-11-07T13:22:36+05:30 INFO [checks-client] Downloading the checks bundle...
2026-11-07T13:22:36+05:30 ERROR [misconfig] Falling back to embedded checks err="failed to download checks bundle: download error: OCI repository error: 1 error occurred:\n\t* error getting credentials - err: exec: \"docker-credential-desktop\": executable file not found in $PATH, out: ``\n\n"
2026-11-07T13:22:36+05:30 INFO [terraform scanner] Scanning root module file_path="."
2026-11-07T13:22:36+05:30 WARN [terraform parser] Variable values were not found in the environment or variable files. Evaluating may not work correctly. module="root" variables="aws_accounts"
2026-11-07T13:22:36+05:30 INFO Detected config files num=1
Report Summary
Target Type Misconfigurations
. terraform 0
Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)
md-sha@MDs-MacBook-Air enterprise-identity-governance %
day-15-2 main* 0 0
Ln 10, Col 1 Spaces: 4 UTF-8 LF Terraform

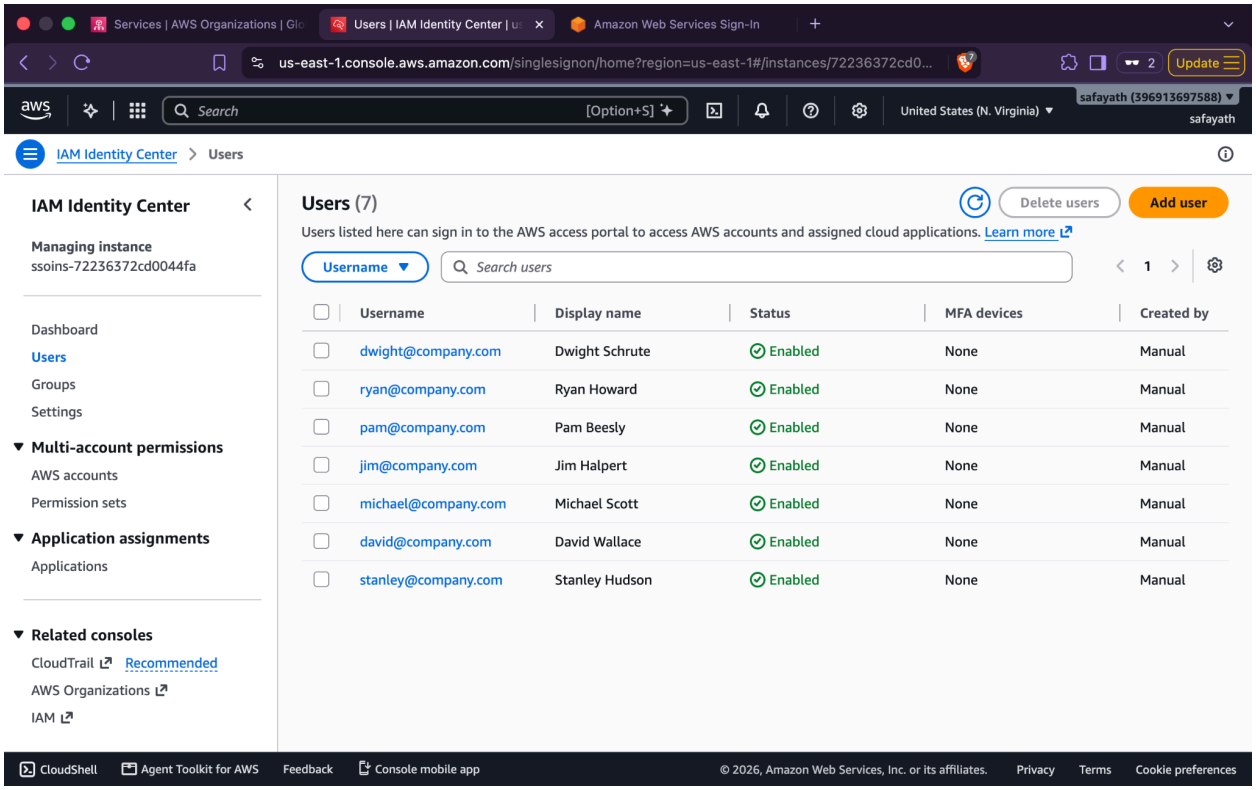

Terroform_projects
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Terroform_projects
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enterprise-identity-gove...
modules
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engineering-policy.json
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.terraform.lock.hcl
backend.tf
dev.tfvars
employees.csv
locals.tf
main.tf
outputs.tf
provider.tf
variables.tf
versions.tf
data.tf
employees.csv
local.tf
main.tf
Outline
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Problems
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Debug Console
Terminal
zsh - enterprise-identity-governance
.tfvars .
md-sha@MDs-MacBook-Air enterprise-identity-governance % trivy config --skip-check-update --tf-vars dev .tfvars .
2026-11-07T13:23:48+05:30 INFO [misconfig] Misconfiguration scanning is enabled
2026-11-07T13:23:48+05:30 INFO [checks-client] No downloadable checks were loaded as --skip-check-update is enabled, loading from existing cache...
2026-11-07T13:23:48+05:30 ERROR [misconfig] Falling back to embedded checks err="failed to check cache: cache does not exist at \"/Users/md-sha/Library/Caches/trivy/policy/content\""
2026-11-07T13:23:49+05:30 INFO [terraform scanner] Scanning root module file_path="."
2026-11-07T13:23:49+05:30 INFO Detected config files num=1
Report Summary
Target Type Misconfigurations
. terraform 0
Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)
md-sha@MDs-MacBook-Air enterprise-identity-governance % trivy fs --scanners secret .
2026-11-07T13:23:57+05:30 INFO [secret] Secret scanning is enabled
2026-11-07T13:23:57+05:30 INFO [secret] Please see https://trivy.dev/docs/v0.76/guide/scanner/secret#recommendation for faster secret detection
2026-11-07T13:23:57+05:30 INFO Number of language-specific files num=0
2026-11-07T13:23:57+05:30 INFO [report] No issues detected with scanner(s). scanners=[secret]
Report Summary
Target Type Secrets
- - -
Legend:
day-15-2 main* 0 0
Ln 10, Col 1 Spaces: 4 UTF-8 LF Terraform

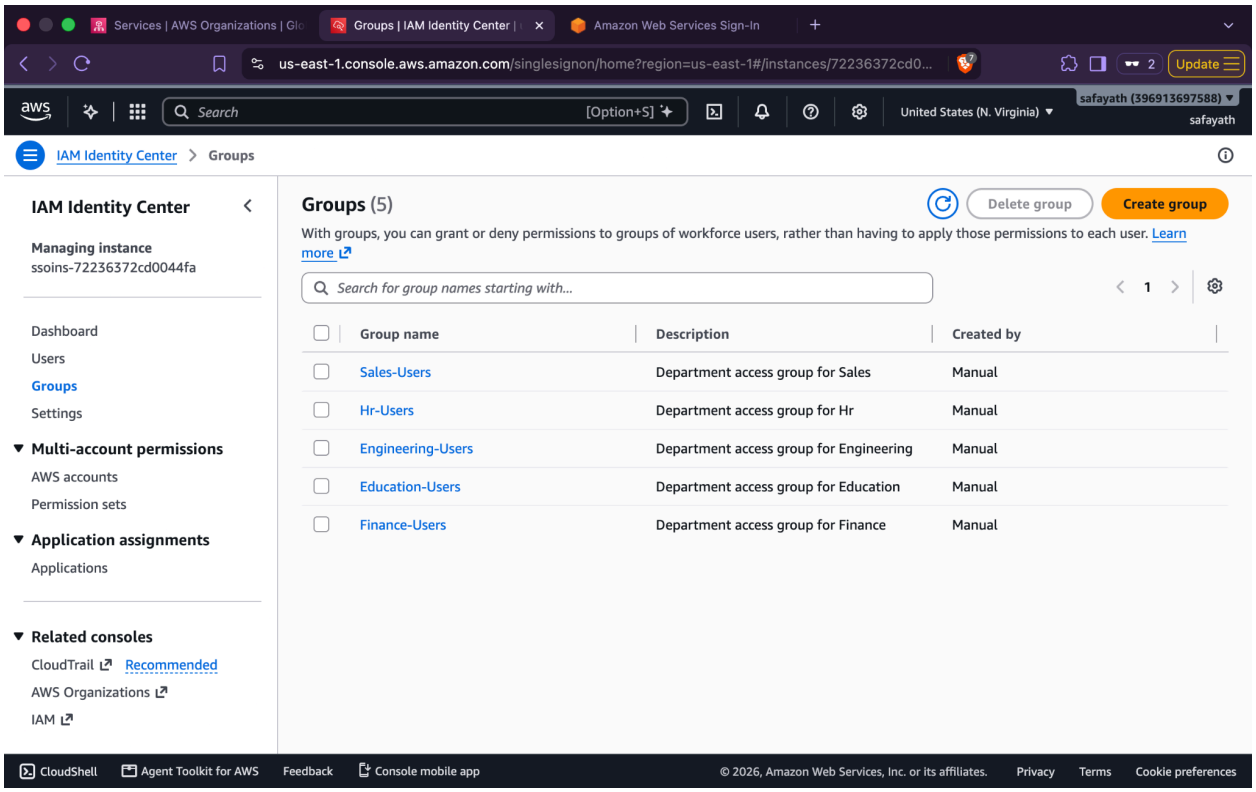

Enterprise AWS Identity Governance
HR Employee CSV
Employee identity and employment status
Terraform Data Processing
CSV decoding · Normalization · Validation · Dynamic maps
AWS IAM Identity Center
Users
Identity provisioning
Groups
Department membership
Permission sets
RBAC policies
Assignments
AWS account access
AWS Development Account
Provisioned IAM roles and workforce access
Security Monitoring
CloudTrail · IAM Access Analyzer · Configuration monitoring

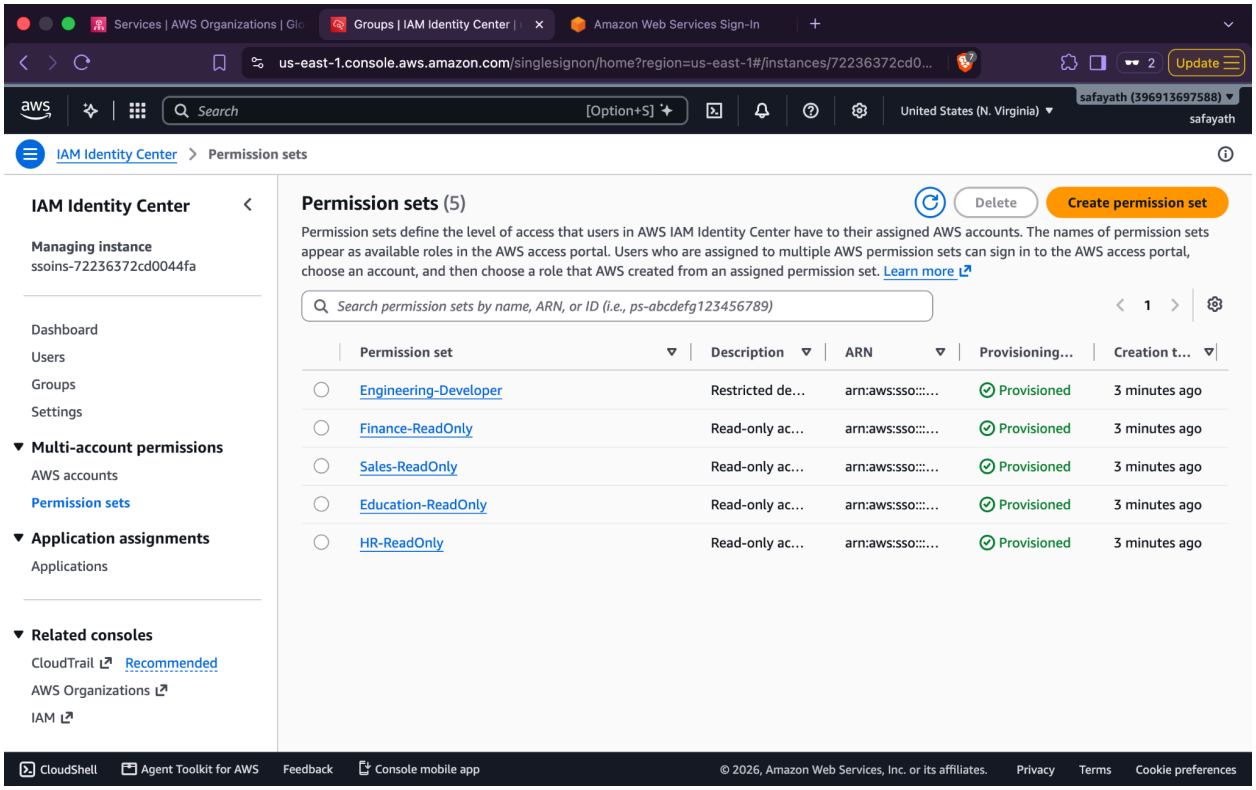

Dashboard | IAM Identity Cent
Amazon Web Services Sign-In
us-east-1.console.aws.amazon.com/singlesignon/home?region=us-east-1#/instances/72236372cd0...
Update
Search
[Option+S]
United States (N. Virginia)
safayath (396913697588)
safayath
IAM Identity Center > Dashboard
IAM Identity Center
Managing instance
ssoins-72236372cd0044fa
Dashboard
Users
Groups
Settings
Multi-account permissions
AWS accounts
Permission sets
Application assignments
Applications
Related consoles
CloudTrail Recommended
AWS Organizations
IAM
You have successfully created the organization instance of IAM Identity Center 72236372cd0044fa.
Dashboard
IAM Identity Center enables you to manage workforce user access to multiple AWS accounts and applications. Learn more
Central management
Prevent account instances
Use service control policies (SCPs) to prevent instances of IAM Identity Center from being created, or isolate the member accounts that are allowed to create account instances.
Learn more about service control policies
IAM Identity Center allows member accounts of an organization to enable AWS applications that are independent of the organization instance with self-managed, account instances of IAM Identity Center. Learn more about account instances
Monitor activities in your instances of IAM Identity Center
With AWS CloudTrail, you can monitor and audit activity in your organization instance and account instances of IAM Identity Center.
Learn about monitoring IAM Identity Center
Settings summary
Go to settings
Specify a unique name for your instance.
Instance name - Edit
-
Identity source
Identity Center directory
Primary Region
us-east-1
Organization ID
o-89vkvcyfe0
Issuer URL (for OIDC applications
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

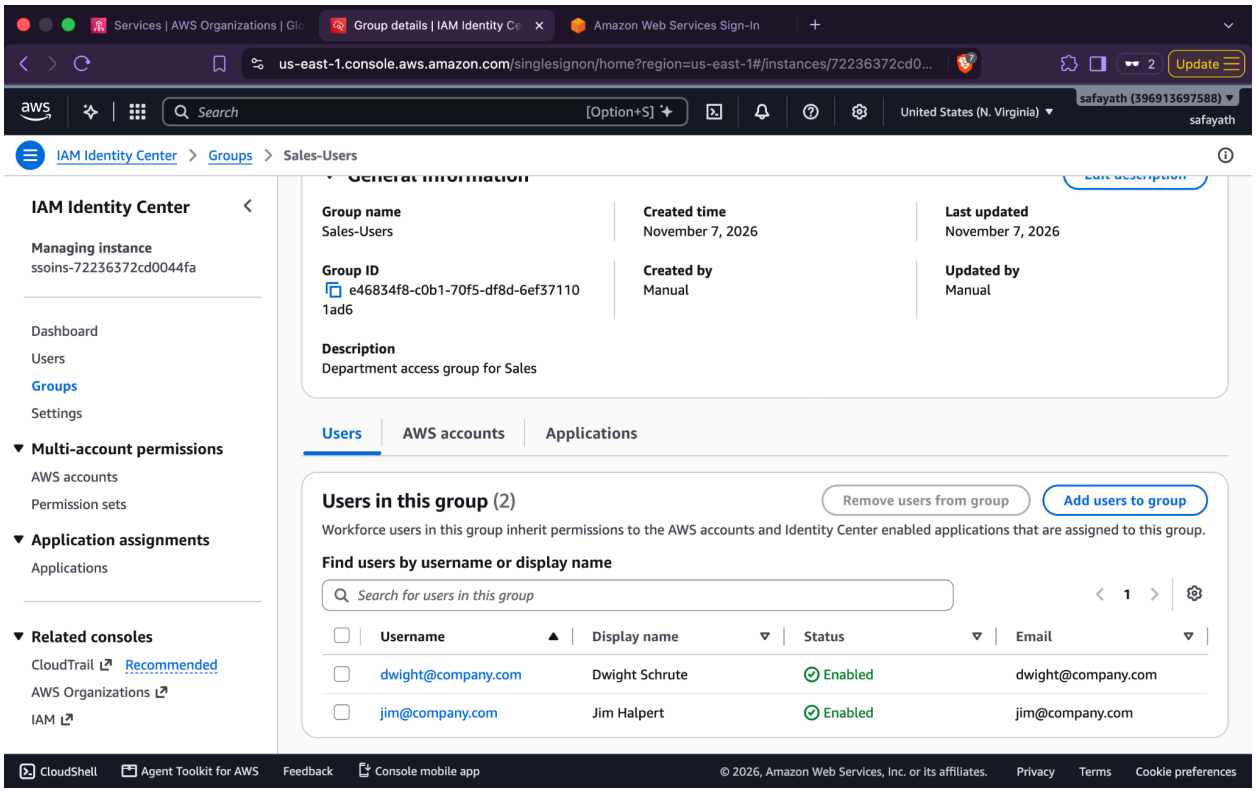
Resource	Count
Identity Center users	7
Department groups	5
Group memberships	7
Permission sets	5
Managed policy attachments	4
Inline policy attachments	1
AWS account assignments	5
Validation resource	1
Total	35

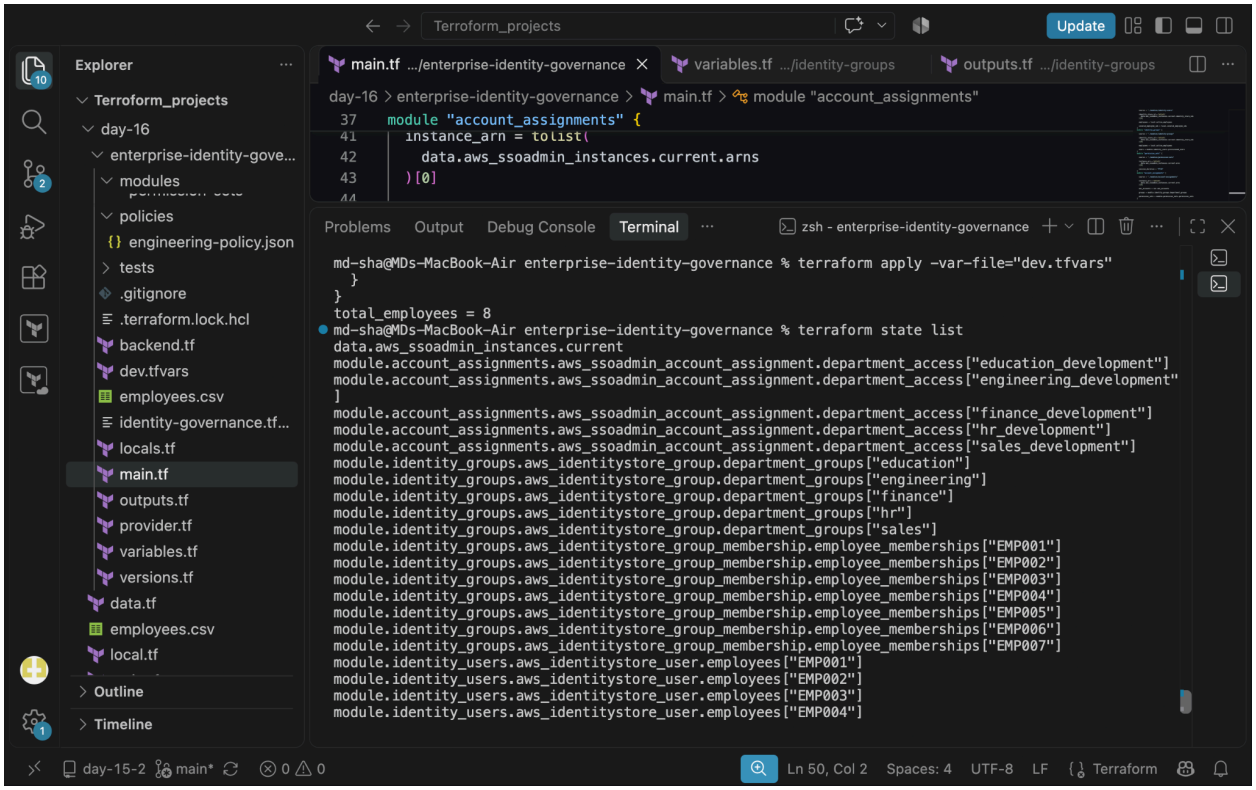



Services | AWS Organizations | Glo
Users | IAM Identity Center | u
Amazon Web Services Sign-In
us-east-1.console.aws.amazon.com/singlesignon/home?region=us-east-1#/instances/72236372cd0...
Update
Search
[Option+S]
United States (N. Virginia)
safayath (396913697588)
safayath
IAM Identity Center > Users
IAM Identity Center
Managing instance
ssoins-72236372cd0044fa
Dashboard
Users
Groups
Settings
Multi-account permissions
AWS accounts
Permission sets
Application assignments
Applications
Related consoles
CloudTrail Recommended
AWS Organizations
IAM
Users (7)
Delete users
Add user
Users listed here can sign in to the AWS access portal to access AWS accounts and assigned cloud applications. Learn more
Username
Search users
1
Username | Display name | Status | MFA devices | Created by
dwight@company.com | Dwight Schrute | Enabled | None | Manual
ryan@company.com | Ryan Howard | Enabled | None | Manual
pam@company.com | Pam Beesly | Enabled | None | Manual
jim@company.com | Jim Halpert | Enabled | None | Manual
michael@company.com | Michael Scott | Enabled | None | Manual
david@company.com | David Wallace | Enabled | None | Manual
stanley@company.com | Stanley Hudson | Enabled | None | Manual
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Services | AWS Organizations | Glo
Groups | IAM Identity Center |
Amazon Web Services Sign-In
us-east-1.console.aws.amazon.com/singlesignon/home?region=us-east-1#/instances/72236372cd0...
Update
Search
[Option+S]
United States (N. Virginia)
safayath (396913697588)
safayath
IAM Identity Center > Groups
IAM Identity Center
Managing instance
ssoins-72236372cd0044fa
Dashboard
Users
Groups
Settings
Multi-account permissions
AWS accounts
Permission sets
Application assignments
Applications
Related consoles
CloudTrail Recommended
AWS Organizations
IAM
Groups (5)
Delete group
Create group
With groups, you can grant or deny permissions to groups of workforce users, rather than having to apply those permissions to each user. Learn more
Search for group names starting with...
1
Group name | Description | Created by
Sales-Users | Department access group for Sales | Manual
Hr-Users | Department access group for Hr | Manual
Engineering-Users | Department access group for Engineering | Manual
Education-Users | Department access group for Education | Manual
Finance-Users | Department access group for Finance | Manual
CloudShell
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Feedback
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Services | AWS Organizations | Glo
Groups | IAM Identity Center |
Amazon Web Services Sign-In
us-east-1.console.aws.amazon.com/singlesignon/home?region=us-east-1#/instances/72236372cd0...
Update
Search
[Option+S]
United States (N. Virginia)
safayath (396913697588)
safayath
IAM Identity Center > Permission sets
IAM Identity Center
Managing instance
ssoins-72236372cd0044fa
Dashboard
Users
Groups
Settings
Multi-account permissions
AWS accounts
Permission sets
Application assignments
Applications
Related consoles
CloudTrail Recommended
AWS Organizations
IAM
Permission sets (5)
Delete
Create permission set
Permission sets define the level of access that users in AWS IAM Identity Center have to their assigned AWS accounts. The names of permission sets appear as available roles in the AWS access portal. Users who are assigned to multiple AWS permission sets can sign in to the AWS access portal, choose an account, and then choose a role that AWS created from an assigned permission set. Learn more
Search permission sets by name, ARN, or ID (i.e., ps-abcdefg123456789)
Permission set
Description
ARN
Provisioning...
Creation t...
Engineering-Developer
Restricted de...
arn:aws:sso:::...
Provisioned
3 minutes ago
Finance-ReadOnly
Read-only ac...
arn:aws:sso:::...
Provisioned
3 minutes ago
Sales-ReadOnly
Read-only ac...
arn:aws:sso:::...
Provisioned
3 minutes ago
Education-ReadOnly
Read-only ac...
arn:aws:sso:::...
Provisioned
3 minutes ago
HR-ReadOnly
Read-only ac...
arn:aws:sso:::...
Provisioned
3 minutes ago
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Services | AWS Organizations | Glo
Group details | IAM Identity Ce
Amazon Web Services Sign-In
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Update
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United States (N. Virginia)
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IAM Identity Center > Groups > Sales-Users
IAM Identity Center
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Dashboard
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AWS Organizations
IAM
Group name
Sales-Users
Created time
November 7, 2026
Last updated
November 7, 2026
Group ID
e46834f8-c0b1-70f5-df8d-6ef371101ad6
Created by
Manual
Updated by
Manual
Description
Department access group for Sales
Users
AWS accounts
Applications
Users in this group (2)
Remove users from group
Add users to group
Workforce users in this group inherit permissions to the AWS accounts and Identity Center enabled applications that are assigned to this group.
Find users by username or display name
Search for users in this group
Username
Display name
Status
Email
dwight@company.com
Dwight Schrute
Enabled
dwight@company.com
jim@company.com
Jim Halpert
Enabled
jim@company.com
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Terroform_projects
Update
Explorer
Terroform_projects
day-16
enterprise-identity-gove...
modules
policies
engineering-policy.json
tests
.gitignore
.terraform.lock.hcl
backend.tf
dev.tfvars
employees.csv
identity-governance.tf...
locals.tf
main.tf
outputs.tf
provider.tf
variables.tf
versions.tf
data.tf
employees.csv
local.tf
Outline
Timeline
main.tf .../enterprise-identity-governance
variables.tf .../identity-groups
outputs.tf .../identity-groups
day-16 > enterprise-identity-governance > main.tf > module "account_assignments"
37 module "account_assignments" {
41 instance_arn = tolist(
42 data.aws_ssoadmin_instances.current.arns
43)[0]
Problems
Output
Debug Console
Terminal
zsh - enterprise-identity-governance
md-sha@MDs-MacBook-Air enterprise-identity-governance % terraform apply -var-file="dev.tfvars"
}
}
total_employees = 8
md-sha@MDs-MacBook-Air enterprise-identity-governance % terraform state list
data.aws_ssoadmin_instances.current
module.account_assignments.aws_ssoadmin_account_assignment.department_access["education_development"]
module.account_assignments.aws_ssoadmin_account_assignment.department_access["engineering_development"
]
module.account_assignments.aws_ssoadmin_account_assignment.department_access["finance_development"]
module.account_assignments.aws_ssoadmin_account_assignment.department_access["hr_development"]
module.account_assignments.aws_ssoadmin_account_assignment.department_access["sales_development"]
module.identity_groups.aws_identitystore_group.department_groups["education"]
module.identity_groups.aws_identitystore_group.department_groups["engineering"]
module.identity_groups.aws_identitystore_group.department_groups["finance"]
module.identity_groups.aws_identitystore_group.department_groups["hr"]
module.identity_groups.aws_identitystore_group.department_groups["sales"]
module.identity_groups.aws_identitystore_group_membership.employee_memberships["EMP001"]
module.identity_groups.aws_identitystore_group_membership.employee_memberships["EMP002"]
module.identity_groups.aws_identitystore_group_membership.employee_memberships["EMP003"]
module.identity_groups.aws_identitystore_group_membership.employee_memberships["EMP004"]
module.identity_groups.aws_identitystore_group_membership.employee_memberships["EMP005"]
module.identity_groups.aws_identitystore_group_membership.employee_memberships["EMP006"]
module.identity_groups.aws_identitystore_group_membership.employee_memberships["EMP007"]
module.identity_users.aws_identitystore_user.employees["EMP001"]
module.identity_users.aws_identitystore_user.employees["EMP002"]
module.identity_users.aws_identitystore_user.employees["EMP003"]
module.identity_users.aws_identitystore_user.employees["EMP004"]
day-15-2
main*
0 0
Ln 50, Col 2
Spaces: 4
UTF-8
LF
Terraform